

POLITECNICO
MILANO 1863

Best Bike Paths – DD Document

DD document for the
Software Engineering 2 course assignment

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1 Introduction

1.1. Purpose

The purpose of this document is to serve as a detailed descriptor of the BBP project, to be used by actual developers and as an actual agreement between customer and contractors: it will contain detailed informations about the architecture of the system, its full functionalities, constraints and it will allow the customer to validate the integrity and completeness of said system. With this DD, the customer will also be able to validate all of the requirements, to verify if the system finally meets the expectations.

1.2. Scope

The Best Bike Paths (BBP) project, initiated by a leading bike association, aims to be an advanced support system for the cycling community designed to thrive within a transport landscape dominated by cars.

The core idea is to foster a collaborative digital environment where cyclists can actively record, consult, and publish both their own and other users' riding paths.

This platform will serve as the definitive centralized resource for finding highly recommended routes tailored precisely to the user's specific experience, preferences, and safety considerations.

Ultimately, the system aims to significantly enhance the quality of the cycling experience by providing verified, community-rated routes, thereby connecting riders and making high-quality path discovery universally accessible

1.3. Definitions, Acronyms, Abbreviations

- BBP: Best Bike Paths
- RASD: Requirements Analysis and Specification Document
- DD: Design Document

1.4. Reference Documents

- Specification document assignment
- IEEE standard documentation for DD
- Materials for the Software Engineering 2 course made available on WeBeep

1.5. Document Structure

The document is organized in seven different chapters that will cover every detail on the actual design of our project:

- **Chapter 1 - Introduction:** this chapter shows the aim of the doucment, what it is needed to know before to actually start the read of the whole text and restates the scope of the project.
- **Chapter 2 - Architectural Design:** this chapter shows all of the main components of the whole system, including architectural choices and patterns used.
- **Chapter 3 - User Interface Design:** this chapter shows and explains how the mockup pages we created integrate all of the implemented functionalities.
- **Chapter 4 - Requirements Traceability:** this chapter shows how we ensured that every requirement needed to the system to work is being satisfied with relative design choices.
- **Chapter 5 - Implementation, Integration and Test Plan:** this chapter shows how all of the components are implemented and their integration inside the system, also providing a test plan for all of their instances.
- **Chapter 6 - Effort Spent:** this chapter includes a table where the amount of hour of commitment of every participant is shown relative to its section.
- **Chapter 7 - References:** finally, the last chapter contains all of our document references used to better refine our DD.

2 Architectural Design

2.1. Overview

2.2. Component View

2.2.1. Mobile

2.2.2. Backend

2.3. Deployment View

2.4. Runtime View

2.5. Component Interfaces

2.6. Styles and Patterns

2.7. Other Design Decisions

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5.1. Overview and Implementation Plan

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